Notes: Opler and Titman (JF, 1994)
Financial Distress and Corporate Performance

Contents

1 Big picture

- **Question.** Does financial distress impair corporate performance, or can leverage discipline managers and improve performance during downturns?
- **Core empirical challenge.** Firms that become financially distressed often do so after bad performance. If one studies firms that later file for bankruptcy, sales losses may be a cause rather than a consequence of distress.
- **Research design.** The paper identifies industries experiencing economic distress and asks whether high-leverage firms in those industries underperform their more conservatively financed industry rivals.
- **Main finding.** Highly leveraged firms lose substantial market share in industry downturns. Firms in leverage deciles 8–10 experience 13.6 percentage points lower industry-adjusted sales growth in distressed industries; firms in leverage decile 10 experience roughly 26.4 percentage points lower sales growth than firms in decile 1.
- **Cost versus benefit of distress.** If leverage forces efficient downsizing, one would expect lower sales but possibly higher operating income or stock value. Instead, highly leveraged firms in distressed industries also have worse stock returns and operating income, supporting costly financial distress.
- **Mechanism evidence.** Losses are larger for firms with R&D and in concentrated industries, consistent with customer-driven losses for specialized products and competitor-driven market-share losses.
- **Takeaway.** Financial condition affects product-market performance. High leverage makes firms vulnerable when industry shocks arrive, creating indirect costs of financial distress through customers, competitors, and operating decisions.

2 Incremental contribution of the paper

- **Identification contribution.** The paper shifts the empirical design from "firms that become distressed" to "industries that become distressed." This reduces reverse causality because the shock is defined at the industry level, not by the firm-specific collapse in sales or value.
- **Product-market contribution.** It measures financial distress costs in the product market, especially lost sales and market share, rather than only legal costs, direct bankruptcy costs, or accounting losses.
- **Mechanism contribution.** It distinguishes three explanations for sales losses: customer driven, competitor driven, and manager driven. The R&D, concentration, size, operating-income, and stock-return tests help interpret which mechanism is plausible.
- **Capital-structure contribution.** The paper provides evidence relevant for ex ante leverage choice. Because performance is compared by pre-shock leverage inside industry downturns, the results speak to whether firms should avoid leverage because of potential future distress costs.
- **Literature contribution.** Earlier work showed that distressed firms cut investment, sell assets, or change management. This paper shows that leverage itself is associated with product-market losses during downturns, not merely with internal restructuring after distress occurs.

3 Data and measurement

3.1 Sample and period

- The sample uses COMPUSTAT industrial firms from 1972 to 1991.
- The analysis is organized around base years. Outcomes are measured over a two-year window centered on the base year.

- The final sample contains 46,799 publicly traded firm-years, of which 1,368 are in industries classified as having poor performance.
- Financial firms and other unsuitable observations are removed through the usual COMPUSTAT data filters.

Interpretation: The sample is large enough to compare high- and low-leverage firms within many industries and over many years. This is necessary because the design needs both distressed and normal industries.

3.2 Industry distress measure

A 3-digit SIC industry is classified as distressed when it satisfies both conditions over the two-year period centered on the base year:

1. median industry sales growth is negative;
2. median industry stock return is below -30%.

Interpretation: The two-part definition is important. Negative sales growth alone could capture mild cyclical changes. A large negative stock return indicates that the market views the downturn as serious and unexpected.

3.3 Leverage measure

Leverage is measured two years before the base year:

$$\text{Debt/Assets}_{i,t-2} = \frac{\text{long-term debt} + \text{short-term debt}}{\text{total assets}}.$$

The paper uses book values rather than market values to reduce the concern that market leverage mechanically forecasts future sales and returns through market-value changes.

High leverage is measured using dummy variables:

- top three leverage deciles, i.e. deciles 8–10;
- top leverage decile, i.e. decile 10.

Interpretation: Leverage is measured before the industry shock so that the analysis compares firms that enter the downturn with different financial conditions.

3.4 Performance outcomes

The main outcomes are industry-adjusted measures:

- two-year sales growth;
- two-year stock return;
- two-year change in operating income/assets;
- asset sales/assets;
- employment growth;
- capital expenditure growth;
- exit outcomes: bankruptcy, merger, and other exits.

Industry adjustment subtracts the 3-digit SIC industry mean outcome from each firm outcome.

Interpretation: Industry adjustment turns the question into a relative-performance question: did high-leverage firms perform worse than their own industry peers during the same industry downturn?

3.5 Mechanism variables

- **R&D intensity:** an R&D/sales ratio above 0.1% two years before the base year. This proxies for specialized products and customer dependence.
- **Industry concentration:** four-firm concentration ratio above 40% in 1981. This proxies for strategic interaction and possible predatory or aggressive competitor responses.
- **Firm size:** sales greater or less than \$100 million. This tests whether large or small firms are more vulnerable to distress costs.

Interpretation: The heterogeneity variables map directly to the theory. Customer-driven losses should be strongest in specialized-product firms. Competitor-driven losses should be strongest in concentrated industries.

4 Empirical specifications

4.1 Baseline regression

The main regression is:

$$\begin{aligned} Performance_i = \alpha &+ \beta_1 \log(Sales_i) + \beta_2 IA\,Profitability_i + \beta_3 IA\,Investment_i \\ &+ \beta_4 IA\,AssetSales_i + \beta_5 DistressedIndustry_i + \beta_6 HighLeverage_i \\ &+ \beta_7 (DistressedIndustry_i \times HighLeverage_i) + \varepsilon_i . \end{aligned}$$

- $Performance_i$ is industry-adjusted sales growth, stock return, or operating income change.
- IA denotes industry adjusted.
- $HighLeverage_i$ is either deciles 8–10 or decile 10.
- β_7 is the main coefficient of interest.

Interpretation: β_7 measures whether leverage is especially harmful during industry distress, over and above any general relation between leverage and performance.

4.2 Heterogeneity specifications

The paper re-estimates the baseline regression separately by:

- R&D-intensive versus non-R&D-intensive firms;
- high-concentration versus other industries;
- small versus large firms.

Interpretation: These split-sample tests evaluate mechanisms. Larger losses in R&D firms point toward customer-driven distress costs. Larger losses in concentrated industries point toward competitor-driven distress costs.

4.3 Manager-driven tests

The paper studies asset sales, employment growth, and capital expenditure growth:

$$Outcome_i = \alpha + \beta_1 \log(Sales_i) + \cdots + \beta_7(Distressed_i \times HighLeverage_i) + \varepsilon_i.$$

Interpretation: If the observed sales decline reflects efficient manager-driven downsizing, the interaction should predict higher asset sales or lower inefficient spending in a way that improves performance. The results provide only limited support for the manager-driven story.

4.4 Exit regressions and exit rates

The paper also examines whether high-leverage firms in distressed industries disappear from the sample through bankruptcy, merger, or other reasons.

Interpretation: Exit tests are important because survivorship bias can understate financial distress costs. If the worst high-leverage firms fail and drop from performance regressions, measured sales and profit losses among survivors are conservative.

5 Identification strategy

5.1 Why industry downturns reduce reverse causality

Previous work that studies firms close to bankruptcy faces a severe causality problem: sales declines may cause financial distress rather than result from it. Opler and Titman instead start from industry-wide downturns and compare high- and low-leverage firms in the same industry environment.

Interpretation: The design does not make leverage random, but it shifts the shock from the firm level to the industry level. This makes it more plausible that observed relative underperformance reflects the interaction of leverage with industry distress.

5.2 Role of pre-shock leverage

Leverage is measured two years before the base year, before the performance window. This reduces mechanical simultaneity between current performance and measured leverage.

Interpretation: Pre-shock book leverage is intended to capture financial vulnerability before the distress event, not the market reaction to the event.

5.3 Industry-adjusted outcomes

By subtracting 3-digit industry means, the paper compares firms exposed to the same industry shock. This helps remove common demand, input-cost, and macro shocks.

Interpretation: The regression asks: within an industry downturn, do highly leveraged firms lose more than their less leveraged competitors?

5.4 Controls

The controls include prior size, profitability, investment, and asset sales. These variables address the concern that highly leveraged firms might differ systematically in operating performance, growth opportunities, and divestiture behavior before the downturn.

5.5 Remaining limitations

- Leverage is still a choice, not randomly assigned.
- Highly leveraged firms may be less efficient even before the downturn.
- Book leverage reduces but does not eliminate the possibility that leverage proxies for poor prospects.
- The high-leverage distressed-industry group is an imperfect proxy for true financial distress: it includes some firms that are not actually distressed and excludes some distressed firms.

Interpretation: These limitations likely reduce statistical power and may understate financial distress costs if firms with high potential distress costs avoid leverage ex ante.

6 Tables and interpretations

6.1 Table I and Table II

Read: Table I classifies three explanations for why highly leveraged firms may lose sales during downturns: customer driven, competitor driven, and manager driven. Table II shows the annual distribution of distressed and normal industries.

Interpretation: Table I provides the roadmap for the paper's mechanism tests. Table II shows that distressed industries are relatively rare: only 1,368 of 46,799 firm-years are in poor-performing industries.

Economic message: The paper is not simply documenting average leverage-performance relations; it studies leverage when industries are hit by severe negative shocks.

Table I

Summary of Potential Causes of Performance Declines by Highly Leveraged Firms in Periods of Economic Distress

Explanation of Performance Decline	Explanation	Explanation Predicts Loss of Sales Revenue?	Explanation Predicts Decline in Firm Value?	Other Predictions
Customer driven	Customers and stakeholders abandon the firm	Yes	Yes	Performance decline wors for firms with special products
Competitor driven	Competitors reduce prices to gain market share	Yes	Yes	Performance decline wors concentrated industr
Manager driven	Managers efficiently downsize by cutting poorly performing assets	Yes	No	Performance decline may related to firm size

Table I: Mechanisms for performance declines.

Table II

Distribution of Firms by Year

The sample consists of 46,799 publicly traded firm-years in the 1972 to 1991 period. Of these firm years, 1,368 (3 percent) were in industries with poor performance that exhibited negative median sales growth and median stock returns below −30 percent.

Base Year	No. of Firms in Industries with Normal Performance	No. of Firms in Industries with Poor Performance	No. of Poorly Performing Industries
1974	1,973	36	3
1975	2,344	0	0
1976	3,417	9	1
1977	3,237	0	0
1978	2,967	0	0
1979	2,865	0	0
1980	2,754	10	2
1981	2,626	38	6
1982	2,462	228	5
1983	2,320	215	4
1984	2,451	205	6
1985	2,385	198	9
1986	2,473	148	4
1987	2,749	16	3
1988	2,921	9	3
1989	2,960	92	12
1990	2,527	161	14

Table II: Distribution of firms by year.

6.2 Table III: Description of Sample Firms in Industries with Poor Performance and Normal Performance

Read: Table III compares firms in poor-performance industries with firms in normal industries. Poor-industry firms are smaller, have lower sales growth, lower stock returns, and lower operating income changes. They also have lower R&D intensity on average.

Interpretation: The table explains why controls are necessary. Firms in distressed industries differ from firms in normal industries even before mechanism tests are run.

Economic message: Industry downturns are real adverse shocks, not just relabelled normal variation.

Table III

Description of Sample Firms in Industries with Poor Performance and Normal Performance

The sample contains 46,799 years of data in the 1972 to 1991 period. Of these firms, 1,368 (3 percent) were in industries with poor performance with negative median sales growth and median stock returns below −30 percent. Assets are measured at book value. Ex ante leverage is measured two years before the base year. Sales growth, operating income change, and stock returns are measured over a two-year period centered on the base year.

Variable	Mean	Quartile One	Median	Quartile Three
Panel A: Firms in Industries Experiencing Poor Performance (N = 1,368)				
Ex ante debt/assets (%)	29	7.1	26	42
Base-year debt/assets (%)	34	7.1	30	50
Base-year sales ($ millions)	181	1.5	9.9	60
2-year sales growth (%)	−7.4	−41	−15	13
2-year stock return (%)	−39	−75	−50	−18
2-year operating income change (%)	−3.6	−11.5	−3.5	3.6
R&D expense/sales (%)	0.9	0	0	0.01
Four-firm industry concentration (%)	22.3	20.4	20.4	20.4
Panel B: Firms in Industries Experiencing Normal Performance (N = 45,431)				
Ex ante debt/assets (%)	30	12	26	41
Base-year debt/assets (%)	31	12	26	41
Base-year sales ($ millions)	480	9.5	40	173
2-year sales growth (%)	24	−0.3	20	43
2-year stock return (%)	18	−28	8.6	55
2-year operating income change (%)	3.9	−3.0	2.9	9.5
R&D expense/sales (%)	2.4	0	0	1.6
Four-firm concentration ratio (%)	28.4	17.7	23.3	34.9

Table III: Description of sample firms in poor and normal performance industries.

6.3 Table IV: OLS Regressions Predicting Industry-Adjusted Sales Growth, Stock Returns, and Operating Income Growth

Read: The interaction between high leverage and distressed industry is negative for sales growth, stock returns, and operating income. For leverage deciles 8–10, sales growth is 13.6 percentage points lower in distressed industries. For the top leverage decile, the relative sales-growth loss is about 26.4 percentage points.

Interpretation: Table IV is the main result. High leverage is harmful in general, but especially harmful in industry downturns. The negative operating-income result helps rule out the idea that sales losses are simply efficient downsizing.

Economic message: Financial distress has real product-market costs: customers, suppliers, or competitors appear to penalize financially weak firms during downturns.

OLS Regressions Predicting Mean Industry-Adjusted Sales Growth, Stock Returns, and Operating Income Growth in the 1972 to 1991 Period

Industry adjustment is carried out by subtracting the 3-digit SIC industry mean from the firm's performance. Ex ante leverage is measured two years prior to the base year and is defined as the book value of long-term and short-term debt divided by total assets. Stock returns, operating income growth, and sales growth are measured over a two-year period centered on the base year. Operating income is defined as earnings before interest, taxes, and depreciation. Distressed industries had negative mean sales growth and mean stock returns below −30 percent. A binomial sign test is used to measure the significance of the proportion of leveraged firms in distressed industries with above-median industry performance compared to the same proportion for leveraged firms in nondistressed industries.

	Industry-Adjusted Sales Growth		Industry-Adjusted Stock Returns		Industry-Adjusted Operating Income	
	Whole Sample	Leverage Deciles 1 and 10 Only	Whole Sample	Leverage Deciles 1 and 10 Only	Whole Sample	Leverage Deciles 1 and 10 Only
Intercept	−0.111	−0.110	−0.119	−0.155	−0.025	−0.043
	(−26.1)***	(−11.4)***	(−16.5)***	(−9.88)***	(−3.85)***	(−3.00)**
Log of sales	0.027	0.052	0.031	0.046	0.0056	0.0096
	(31.4)***	(22.0)***	(21.1)***	(11.8)***	(4.23)***	(2.84)***
Industry-adjusted operating income/assets before base year	−0.011	−0.063	0.090	0.019	−0.180	−0.096
	(−1.39)	(−5.39)***	(6.23)***	(0.94)	(−15.0)***	(−5.95)***
Industry-adjusted investment/assets before base year	0.411	0.299	−0.109	−0.190	0.244	0.153
	(19.9)***	(7.31)***	(−3.34)**	(−2.91)***	(7.65)***	(2.57)***
Industry-adjusted asset sales/assets	−0.552	−0.343	−0.364	−0.104	−0.024	−0.014
	(−23.2)***	(−7.64)***	(−8.732)	(−1.32)	(−0.59)	(−0.20)
Distressed industry dummy	0.111	0.206	0.104	0.221	−0.0045	0.107
	(8.00)***	(6.08)***	(5.74)***	(5.42)***	(−0.71)	(2.05)**
Leverage deciles 8–10 dummy	−0.030	—	−0.040	—	−0.0045	—
	(−7.01)***		(−5.92)***		(−0.71)	
Distressed industry dummy x leverage deciles 8–10 dummy	−0.136	—	−0.119	—	−0.067	—
	(−5.64)***		(−3.81)***		(−1.72)*	

Table IV: Main regressions, part 1.

Table IV—*Continued*

	Industry-Adjusted Sales Growth		Industry-Adjusted Stock Returns		Industry-Adjusted Operating Income	
	Whole Sample	Leverage Deciles 1 and 10 Only	Whole Sample	Leverage Deciles 1 and 10 Only	Whole Sample	Leverage Deciles 1 and 10 Only
Leverage decile 10 dummy	—	−0.109	—	−0.071	—	−0.007
		(−10.6)***		(−4.50)***		(−0.50)
Distressed industry dummy x leverage decile 10 dummy	—	−0.264	—	−0.265	—	−0.147
		(−5.24)***		(−4.40)***		(1.82)*
Adjusted R^2	0.05	0.07	0.02	0.03	0.01	0.01
No. of observations	46,623	9,394	33,711	6,019	38,112	7,210
Proportion of leveraged firms above median industry performance in nondistressed industries (%)	48.0	45.0	48.1	45.0	55.4	56.7
Proportion of leveraged firms above median industry performance in distressed industries (%)	44.8	39.3**	42.4**	39.3*	46.3***	43.3**

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table IV: Main regressions, part 2.

6.4 Table V: Heterogeneity by R&D Intensity, Industry Concentration, and Firm Size

Read: Panel A shows stronger leverage-distress effects for R&D-intensive firms. Panel B shows stronger effects in concentrated industries. Panel C shows the effects by firm size.

Interpretation: The R&D split supports the customer-driven mechanism: specialized-product firms are more vulnerable when customers worry about service, quality, or survival. The concentration split supports the competitor-driven mechanism: rivals can exploit financial weakness more in markets where strategic interaction is stronger.

Economic message: The costs of financial distress are not uniform; they are largest when product-market relationships and competitive interaction make financial weakness visible and valuable to rivals.

Table V

OLS Regressions Predicting Mean Industry-Adjusted Sales Growth and Stock Returns as a Function of Industry Performance, Firm Leverage, and Controls in the 1972 to 1991 Period Stratified by Key Firm and Industry Characteristics

Leverage is defined two years prior to the base year as the book value of long-term and short-term debt divided by total assets. Stock returns are dividend and split adjusted and are measured at calendar year-end one year prior to the base year until one year after the base year. Industries that exhibited poor performance had negative mean sales growth and mean stock returns less than −30 percent in the two-year period centered on the base year. R&D-intensive firms are those with R&D/sales two years before the base year above 0.1 percent. High concentration industries are those with a four-firm concentration ratio above 40 percent in 1981. Firm size (sales) is measured one year before the base year.

Panel A: Split by R&D Intensity

	Industry-Adjusted Sales Growth		Industry-Adjusted Stock Returns	
	R&D/Sales > 0.1%	R&D/Sales < 0.1%	R&D/Sales > 0.1%	R&D/Sales < 0.1%
Intercept	−0.097	−0.120	−0.140	−0.017
	(−13.6)***	(−22.9)***	(−12.4)***	(−11.4)***
Log of sales	0.025	0.028	0.036	0.094
	(16.5)***	(27.1)***	(14.9)***	(4.82)***
Industry-adjusted operating income/assets before base year	−0.024	0.018	0.056	0.142
	(−2.16)**	(1.39)	(2.79)***	(6.36)***
Industry-adjusted investment/assets before base year	0.464	0.393	−0.115	−0.111
	(10.2)***	(17.3)***	(−1.70)*	(−3.00)***
Industry-adjusted asset sales/assets	−0.630	−0.519	−0.470	−0.323
	(−13.0)***	(−19.3)***	(−5.72)***	(−6.81)***
Distressed industry dummy	0.132	0.113	0.121	0.094
	(3.67)***	(7.70)***	(2.65)***	(4.82)***
Leverage deciles 8–10 dummy	−0.038	−0.024	−0.0076	−0.049
	(−4.56)***	(−4.78)***	(−0.59)	(−6.19)***
Distressed industry dummy x leverage deciles 8–10 dummy	−0.177	−0.136	−0.198	−0.100
	(−2.47)**	(−5.47)***	(−2.18)**	(−3.07)***
No. of observations	17,701	28,921	13,804	19,906
Adjusted R^2	0.03	0.05	0.02	0.02

Panel B: Split by Industry Concentration

	Industry-Adjusted Sales Growth		Industry-Adjusted Stock Returns	
	High Concentration	Other Industries	High Concentration	Other Industries
Intercept	−0.070	−0.119	−0.095	−0.123
	(−6.61)***	(−24.9)***	(−4.75)***	(−15.4)***
Log of sales	0.013	0.031	0.024	0.032
	(7.12)***	(30.2)***	(6.76)***	(19.4)***
Industry-adjusted operating income/assets before	0.029	−0.031	0.047	0.106
	(1.85)*	(−3.24)***	(1.90)*	(5.81)***

Table V: Heterogeneity, part 1.

Table V—*Continued*

Panel B: Split by Industry Concentration

	Industry-Adjusted Sales Growth		Industry-Adjusted Stock Returns	
	High Concentration	Other Industries	High Concentration	Other Industries
Industry-adjusted investment/assets before base year	0.473	0.402	−0.190	−0.083
	(9.49)***	(17.2)***	(−2.32)**	(−2.25)**
Industry-adjusted asset sales/assets	−0.527	−0.546	−0.485	−0.352
	(−8.58)***	(−20.4)***	(−4.45)***	(−7.56)***
Distressed industry dummy	0.075	0.118	0.058	0.111
	(1.85)*	(7.82)***	(1.04)	(5.66)***
Leverage deciles 8–10 dummy	−0.0025	−0.039	−0.069	−0.037
	(−0.26)	(−8.02)***	(−3.92)***	(−4.94)***
Distressed industry dummy x leverage deciles 8–10 dummy	−0.186	−0.129	−0.134	−0.120
	(−2.30)***	(−4.95)***	(−1.20)	(−3.59)***
No. of observations	7,391	37,360	4,605	27,863
Adjusted R^2	0.03	0.05	0.02	0.02

Panel C: Split by Firm Size

	Industry-Adjusted Sales Growth		Industry-Adjusted Stock Returns	
	Sales < \$100 Million	Sales < \$100 Million	Sales < \$100 Million	Sales > \$100 Million
Intercept	−0.228	−0.276	−0.213	−0.136
	(−40.0)***	(−23.3)***	(−21.0)***	(−6.32)***
Log of sales	0.087	0.038	0.070	0.026
	(50.9)***	(21.2)***	(23.9)***	(7.73)***
Industry-adjusted operating income/assets before base year	−0.093	0.423	0.019	0.370
	(−10.2)***	(14.9)***	(1.17)	(7.49)***
Investment/assets before base year	0.328	0.548	−0.127	−0.132
	(13.3)***	(15.0)***	(−3.27)***	(−2.04)**
Industry-adjusted asset sales/assets	−0.470	−0.573	−0.307	−0.382
	(−16.8)***	(−12.0)***	(−6.27)***	(−4.42)***
Distressed industry dummy	0.187	0.050	0.153	0.132
	(11.2)***	(2.08)**	(7.17)***	(3.42)***
Leverage deciles 8–10 dummy	−0.047	0.013	−0.058	−0.0081
	(−8.22)***	(2.37)**	(−6.33)***	(−0.84)
Distressed industry x leverage deciles 8–10 dummy	−0.163	−0.112	−0.129	−0.162
	(−5.33)***	(−3.35)***	(−3.35)***	(−2.84)***
No. of observations	30,089	16,533	20,387	13,323
Adjusted R^2	0.10	0.07	0.04	0.01

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Table V: Heterogeneity, part 2.

6.5 Table VI and Table VII

Read: Table VI shows mean debt/assets ratios by subgroup. The leverage differences across R&D, concentration, and size splits are not large enough to explain the Table V patterns mechanically. Table VII studies asset sales, employment growth, and capital expenditure growth.

Interpretation: Table VI checks that heterogeneity results are not driven by different leverage distributions. Table VII provides limited support for manager-driven downsizing: employment falls more for high-leverage firms in distressed industries, but asset sales and investment do not strongly support efficient restructuring as the main story.

Economic message: The mechanism evidence points more toward product-market losses than toward pure efficiency-improving internal discipline.

Table VI

Mean Debt / Assets Ratio by Industry and Firm Characteristics Identified in the 1972 to 1991 Period

The number of firms in each cell is shown in parentheses. Leverage is defined two years prior to the base year as the book value of long-term and short-term debt divided by total assets. High-debt firms have leverage in deciles 8 to 10. Stock returns are dividend and split adjusted and are measured at calendar year-end one year prior to the base year until one year after the base year. Industries that exhibited poor performance had negative mean sales growth and mean stock returns less than −30 percent in the two-year period centered on the base year. R&D-intensive firms are those with R&D/sales two years before the base year above 0.1 percent. High concentration industries are those with a four-firm concentration ratio above 40 percent in 1981. Firm size (sales) is measured one year before the base year. Asset sales/assets is the sum of asset sales in the base year and the year after the base year divided by assets.

	Poor Industry Performance		Normal Industry Performance	
	High Debt	Others	High Debt	Others
Panel A: Split by R&D Intensity				
R&D-intensive firms (%)	45.2	14.7	48.4	15.5
	(52)	(157)	(3,627)	(13,892)
Other firms (%)	50.7	13.6	49.4	19.3
	(399)	(760)	(10,297)	(17,615)
Panel B: Split by Industry Concentration				
High concentration (%)	47.1	10.3	49.2	19.9
	(21)	(59)	(2,309)	5,018)
Other industries (%)	49.6	14.2	49.0	17.1
	(412)	(831)	(10,992)	(25,252)
Panel C: Split by Firm Size				
Sales < \$100 million (%)	49.5	11.8	51.2	14.9
	(314)	(772)	(9,259)	(19,877)
Sales > \$100 million (%)	49.6	24.2	46.1	21.0

Table VI: Mean debt/assets by subgroup.

Table VII

OLS Regressions Predicting Mean Industry-Adjusted Asset Sales / Assets, Employment Growth, and Capital Expenditure Growth in the 1972 to 1991 Period

Industry adjustment is carried out by subtracting the 3-digit SIC industry mean from the firm's performance. Ex ante leverage is measured two years prior to the base year and is defined as the book value of long-term and short-term debt divided by total assets. Employment growth and investment (capital expenditures) growth are measured over a two-year period centered on the base year. Distressed industries had negative mean sales growth and mean stock returns below −30 percent. Asset sales/assets is the sum of asset sales in the base year and the year after the base year divided by total assets at book.

Independent Variable	Asset Sales/Assets	Employment Growth	Investment Growth
Intercept	−0.009	−0.188	−0.180
	(−11.0)***	(−25.4)***	(−23.8)***
Log of sales	−0.0032	0.047	0.046
	(19.4)***	(31.7)***	(30.4)***
Industry-adjusted operating income/assets before base year	−0.0081	0.119	0.089
	(−5.18)***	(7.28)***	(5.94)***
Industry-adjusted investment/assets before base year	0.062	−0.199	−0.179
	(15.5)***	(−5.59)***	(−4.88)**
Industry-adjusted asset sales/assets	—	−0.526	−0.513
		(−12.2)***	(−11.6)***
Distressed industry dummy	−0.0081	0.127	0.117
	(−3.02)***	(5.59)***	(5.01)***
Leverage deciles 8–10 dummy	0.012	−0.032	−0.040
	(14.8)***	(−4.50)***	(−5.38)***
Distressed industry dummy x leverage deciles 8–10 dummy	0.0071	−0.096	−0.044
	(1.52)	(−2.42)**	(−1.03)
Adjusted R^2	0.02	0.04	0.04
No. of observations	46,623	39,876	39,644

***Significant at the 1 percent level.

Table VII: Asset sales, employment, and capital expenditure growth.

6.6 Table VIII: Mean Bankruptcy and Merger Rate by Leverage and Industry Performance

Read: Highly leveraged firms in distressed industries have a bankruptcy rate of 6.5%, compared with 2.7% for less leveraged firms in distressed industries. Merger rates do not rise comparably for highly leveraged firms in distressed industries.

Interpretation: Table VIII shows that survivorship bias likely understates the measured performance losses: some high-leverage distressed firms fail and disappear before their full performance losses are observed.

Economic message: Financial distress costs include both observed performance declines among survivors and unobserved losses from firms that exit through bankruptcy.

Table VIII

Mean Bankruptcy and Merger Rate by Leverage and Industry Performance in the 1972 to 1991 Period

Leverage is measured two years prior to the base year and is defined as the book value of long-term and short-term debt divided by total assets. The bankruptcy rate is the proportion of all firms which exited the COMPUSTAT because of bankruptcy or liquidation in the base year or the two years following the base year. The merger rate is the proportion of all firms which exited the COMPUSTAT because they were merged into another firm in the base year or the two years following the base year. The mean rate of exit for other reasons is the proportion of all firms which exited the COMPUSTAT but did not go bankrupt or merge into another firm. Industries which exhibited poor performance had negative median sales growth and median stock market returns below −30 percent in the two year period centered on the base year.

	Mean Bankruptcy Rate (%)	Mean Merger Rate (%)	Mean Rate of Exit for Other Reasons (%)
Poor industry			
Leverage deciles 8–10	6.5	12.1	3.4
Leverage deciles 1–7	2.7	11.4	5.6
Normal industry			
Leverage deciles 8–10	3.9	10.0	2.4
Leverage deciles 1–7	2.0	13.9	1.7

Table VIII: Bankruptcy and merger rates by leverage and industry performance.

7 Interpretive synthesis

- **Core result.** High leverage makes firms vulnerable during industry downturns. The effect appears in sales, stock returns, and operating income.
- **Customer mechanism.** R&D-intensive firms suffer more, consistent with customers avoiding distressed firms that sell specialized products or require future servicing.
- **Competitor mechanism.** Firms in concentrated industries suffer more, consistent with rivals exploiting financial weakness.
- **Manager mechanism.** The paper finds some evidence that high-leverage firms cut employment, but the operating-income and stock-return losses imply that the overall effect is costly rather than value-enhancing.
- **Capital-structure implication.** Firms should consider product-market distress costs when choosing debt. The optimal leverage ratio depends not only on taxes and agency benefits but also on the costs of weak financial condition in downturns.

8 Short summary

- The paper studies whether leverage worsens firm performance in industry downturns.
- Distressed industries have negative median sales growth and median stock returns below −30%.
- Leverage is measured two years before the base year using book debt/assets.
- Outcomes are industry-adjusted sales growth, stock returns, operating income, asset sales, employment, investment, and exit.
- High-leverage firms lose market share in downturns.
- The effect is largest for firms in the top leverage decile.
- Stock returns and operating income also fall, implying that sales losses are costly rather than merely efficient downsizing.
- Effects are stronger in R&D-intensive firms and concentrated industries.

- Bankruptcy rates are higher among high-leverage firms in distressed industries.
- The evidence supports significant indirect costs of financial distress.

9 Conclusion

9.1 Core conclusion

Opler and Titman show that financial distress has economically meaningful indirect costs. Highly leveraged firms perform worse than less leveraged rivals during industry downturns, losing sales, stock-market value, and operating income.

9.2 Economic intuition

Financial weakness changes how customers, competitors, and managers behave. Customers may avoid distressed firms, competitors may attack them, and managers may cut employment or investment. The net effect in the data is costly rather than disciplinary.

9.3 Final takeaway

The paper's lasting contribution is to connect capital structure to product-market outcomes. Leverage is not just a financial claim structure; it changes a firm's competitive position when the industry is under stress.